Ejercicio 4.2.6

Monday, September 29, 2025 9:48 PM

1) Considere $A = A^\dagger$; $K = -K^\dagger$; $U^{-1} = U^\dagger$

P y Q dos operadores genéricos:

a)

I. $(P^\dagger)^{-1} = (P^{-1})^\dagger$

$P P^{-1} = I \quad (P P^{-1})^\dagger = I^\dagger \quad P^\dagger (P^{-1})^\dagger = I$

$(P^\dagger)^{-1} P^\dagger (P^{-1})^\dagger = (P P^\dagger)^{-1}$

$I (P^{-1})^\dagger = (P P^\dagger)^{-1}$

$(P^{-1})^\dagger = (P P^\dagger)^{-1}$

II. $(PQ)^{-1} = Q^{-1} P^{-1}$

$(PQ)(Q^{-1} P^{-1}) = P Q Q^{-1} P^{-1} = P I P^{-1} = I$

$(PQ)(PQ)^{-1} = I$

III. Si $[P, Q] = 0 \Rightarrow P (Q)^{-1} = (Q)^{-1} P$

$[P, Q] = PQ - QP$

$0 = PQ - QP$

$PQ = QP$

$P Q Q^{-1} = Q P Q^{-1}$

$P = Q P Q^{-1}$

$Q^{-1} P = Q^{-1} Q P Q^{-1}$

$Q^{-1} P = P Q^{-1}$

IV. $(e^P)^\dagger = e^{P^\dagger}$

$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} P^n \qquad (e^P)^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!} (P^n)^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!} (P^\dagger)^n = e^{P^\dagger}$

V.

$P e^Q P^{-1} = e^{P Q P^{-1}}$

$P e^Q P^{-1} = P \left(\sum_{n=0}^{\infty} \frac{1}{n!} Q^n \right) P^{-1}$

$= \sum_{n=0}^{\infty} \frac{1}{n!} P Q^n P^{-1}$

Inducción matemática

$P Q^n P^{-1} = (P Q P^{-1})^n \rightarrow$ Sup que es verdad

$P Q^{n+1} P^{-1} = P Q^n Q \cdot P^{-1} \qquad I = P^{-1} P$

$= P Q^n I Q \cdot P^{-1}$

$= (P Q^n P^{-1})(P Q \cdot P^{-1}) \quad P Q^n P^{-1} = (P Q P^{-1})^n$

$= (P Q P^{-1})^n (P Q P^{-1})$

$= (P Q P^{-1})^{n+1}$

b) Si A es hermítico $\qquad \tilde{A} = U^{-1} A U$ también será un operador hermítico.

$U^{-1} = U^\dagger$

$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger$

$\tilde{A} = U^{-1} A U \quad \tilde{A}$

$\tilde{A}^\dagger = (U^{-1} A U)^\dagger = U^\dagger A^\dagger (U^{-1})^\dagger$

$\tilde{A}^\dagger = U^\dagger A^\dagger (U^\dagger)^\dagger$

$\tilde{A}^\dagger = U^\dagger A^\dagger U$

$= U^{-1} A U = \tilde{A}$

$(iA)^\dagger = -i A^\dagger \quad A^\dagger = A$

c.

$e^{iA} = \sum_{n=0}^{\infty} \frac{(iA)^n}{n!}$

$(e^{iA})^\dagger = \sum_{n=0}^{\infty} \frac{(iA)^n}{n!} = \sum_{n=0}^{\infty} \frac{-iA^n}{n!} = e^{-iA}$

$e^{iA} \cdot e^{-iA} = e^0 = I$

$(e^{iA} e^{-iA})^\dagger = I$

$(e^{iA})^\dagger (e^{-iA})^\dagger = I$

$(e^{iA})^\dagger (e^{-iA}) = I$

$(e^{iA})^\dagger (e^{-iA})(e^{iA}) = I (e^{iA})^{-1}$ Son iguales.

$(e^{iA})^\dagger = (e^{iA})^{-1}$ Unitario si $U^\dagger = U^{-1}$

d)

$K^\dagger = -K \qquad \tilde{K} = U^{-1} K U$ esto se cumple $\tilde{K} = iA$

$\tilde{K}^\dagger = U^{-1} (-K) U = -U^{-1} K U = -\tilde{K}$

$K^\dagger = (-i) A^\dagger = -iA = -K$

c) $(AB)^\dagger = B^\dagger A^\dagger = BA$

$A^\dagger = A \wedge B^\dagger = B \qquad AB \Leftrightarrow (AB)^\dagger = AB \Rightarrow BA = AB$

AB hermítico $\Leftrightarrow$ A y B conmutan.

f) S es real $(S^T = -S)$; Operador unidad, probar:

I. $(I - S) \wedge (I + S)$

$(I - S)(I + S) = I - S^2 = (I + S)(I - S)$

II. $(I - S)(I + S)$ es simétrico y $M := (I - S)(I + S)^{-1}$ es ortogonal.

$((I - S)(I + S)^{-1})^T = (I + S)^T (I - S)^T = (I + S^T)(I - S^T)$

Como $S^T = -S$ esto da $(I - S)(I + S)$

$M : M^T M = I. \quad M^T$. usando $(X^{-1})^T = (X^T)^{-1}$

$M^T = ((I - S)(I + S)^{-1})^T = ((I + S)^{-1})^T (I - S)^T = (I + S^T)^{-1} (I - S^T)$

$S^T = -S$

$M^T = (I - S)^{-1} (I + S)$

$M^T M = (I - S)^{-1} (I + S)(I - S)(I + S)^{-1}$.

$(I + S)(I - S) = I - S^2 \qquad (I - S)(I + S) = I - S^2$

$M^T M = (I - S)^{-1} (I - S^2)(I + S)^{-1} = (I - S)^{-1} (I - S)(I + S)(I + S)^{-1} = I$

$M \cdot M^T = I$

g) Considere $R = \begin{pmatrix} \cos\theta & \text{sen}\theta \\ -\text{sen}\theta & \cos\theta \end{pmatrix}$ encuentre una expresión para $R = (I - S)(I + S)^{-1}$

$R(I + S) = I - S \Rightarrow R + RS = I - S$

$(R + I) S = I - R \Rightarrow S = (R + I)^{-1} (I - R)$

siempre que $R + I$ sea invertible

Calculemos $I - R \qquad R + I$

$I - R = \begin{pmatrix} 1 - \cos\theta & -\text{sen}\theta \\ \text{sen}\theta & 1 - \cos\theta \end{pmatrix} \qquad R + I = \begin{pmatrix} 1 + \cos\theta & \text{sen}\theta \\ -\text{sen}\theta & 1 + \cos\theta \end{pmatrix}$

$(R + I)^{-1} (I - R)$

$S = \begin{pmatrix} 0 & -\frac{\text{sen}\theta}{1 + \cos\theta} \\ \frac{\text{sen}\theta}{1 + \cos\theta} & 0 \end{pmatrix} = \tan\left(\frac{\theta}{2}\right) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

$\frac{\text{sen}\theta}{1 + \cos\theta} = \tan\left(\frac{\theta}{2}\right)$